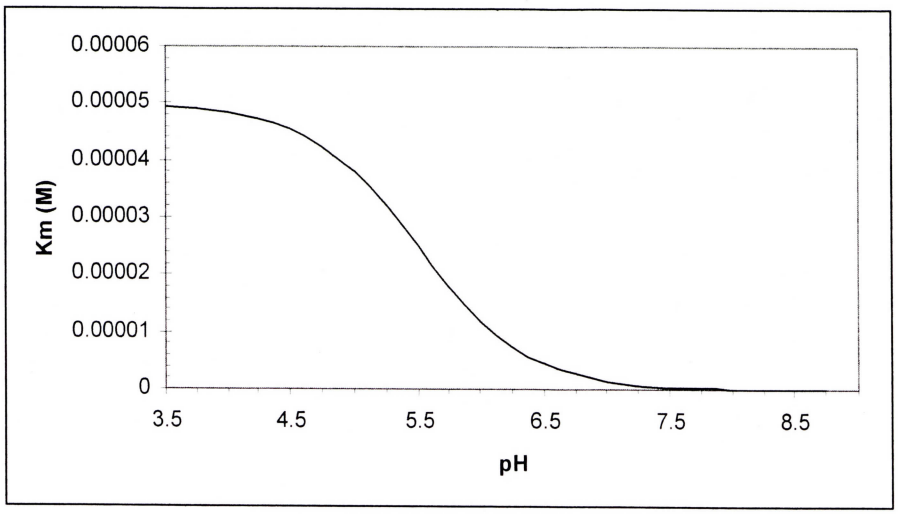
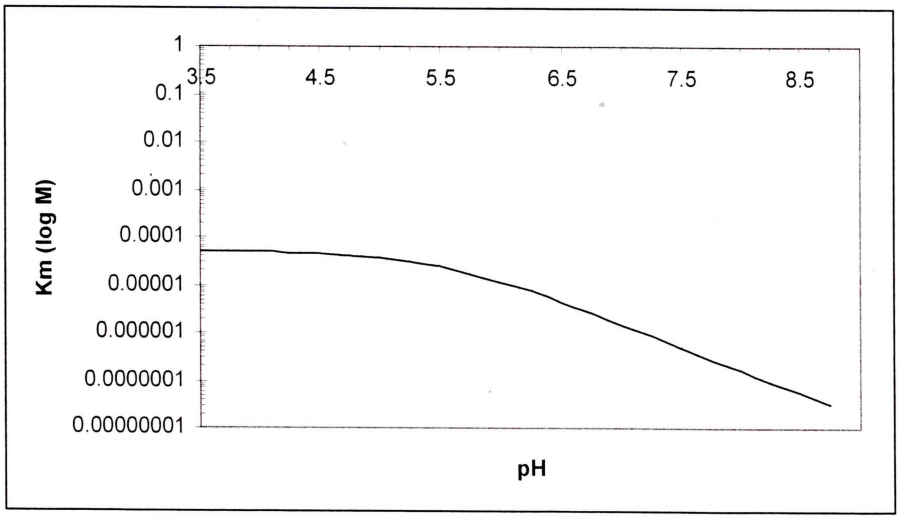
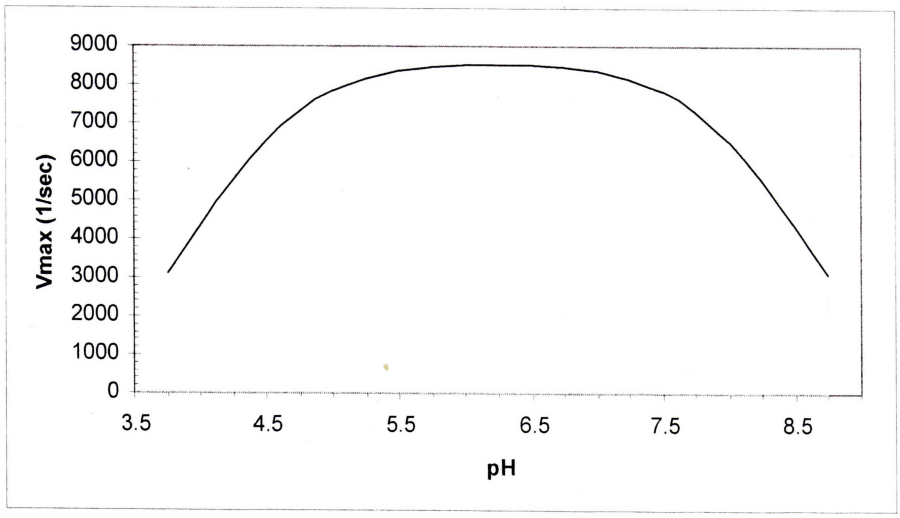
Biochemistry Cumulative Examination 9/7/18

Suppose you are studying the activity of a new enzyme. Other experiments have proven that a glutamic acid, histidine, and a lysine are important in the mechanism. The substrate itself does not have a titratable group.

1. Consider the two graphs (at right and immediately below) showing apparent substrate binding affinity as a function of pH. Based on this data and what you know about the pK_a for each of the three side chains, which of the three side chain(s) do you believe play a role in binding? What is the protonation state(s) of these/this residue(s) in the high affinity state? Explain your reasoning.



2. Consider the graph below showing turnover rate as a function of pH. Based on this data and what you know about the pK_a for each of the three side chains, which of the three side chain(s) do you believe play a role in the enzymatic mechanism? What is the protonation state(s) of these/this residue(s) in the most active state? Explain your reasoning.